

Band-Limited Spectrum of the Inverse Phase Pullback of the Hardy Z -Function

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Abstract

Let ζ denote the Riemann zeta function and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z -function, with ϑ the Riemann–Siegel theta function. Fix a shift $c > c_* := -\inf_{t \in \mathbb{R}} \vartheta'(t)$ and set $\Theta(t) := \vartheta(t) + ct$, a C^∞ bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta' > 0$. Let $Y : \mathbb{R} \rightarrow \mathbb{R}$ be the Omer–Torresani unwarping $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$. The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$, is positive-definite and continuous, and Bochner’s theorem yields a finite non-negative spectral measure μ_Y with $\mu_Y(\mathbb{R}) = 2$. A single integration by parts on the Riemann–Siegel mode expansion, in which the ϑ'' contribution cancels algebraically, gives $\text{supp } \mu_Y \subseteq [-1, 1]$. The Paley–Wiener theorem for functions admitting a compactly supported Bochner spectral measure extends Y to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2} e^{|\Im z|}$; the Akhiezer–Laguerre–Pólya criterion then places Y in the Laguerre–Pólya class, so every complex zero of Y is real. For a nontrivial zero $\rho = \beta + i\gamma$ of ζ , the single complex point $t_\rho := \gamma + i(\frac{1}{2} - \beta)$ satisfies $Z(t_\rho) = 0$ by the functional definition of Z , and the pointwise identity $Z(t_\rho) = \sqrt{\Theta'(t_\rho)} Y(\Theta(t_\rho))$ with $\sqrt{\Theta'(t_\rho)} \neq 0$ yields $Y(\Theta(t_\rho)) = 0$; reality of the zeros of Y gives $\Theta(t_\rho) \in \mathbb{R}$, and the elementary bound $|\Im \Theta(a + ib)| \geq |b|$ implies $\Im t_\rho = 0$, hence $\beta = \frac{1}{2}$.

1 The Shifted Phase and the Unwarped Function

Lemma 1 (Lower bound of ϑ' on $\mathbb{R}$). *$c_* := -\inf_{t \in \mathbb{R}} \vartheta'(t)$ is finite.*

Proof. $\vartheta'(t) = \frac{1}{2} \Re \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi$ is continuous on $\mathbb{R}$ and, by the classical digamma asymptotic applied to $\psi(\frac{1}{4} + \frac{it}{2})$, satisfies $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$ as $|t| \rightarrow \infty$, hence $\vartheta'(t) \rightarrow +\infty$. A continuous function on $\mathbb{R}$ tending to $+\infty$ at infinity is bounded below. $\square$

Definition 2 (Admissible shift and shifted phase). Fix $c > c_*$ and set $\Theta(t) := \vartheta(t) + ct$.

Lemma 3 (Properties of Θ). *For $c > c_*$:*

- (i) $\Theta'(t) = \vartheta'(t) + c > 0$ for every $t \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with C^∞ inverse.
- (ii) ϑ' extends holomorphically to the open strip $\{|\Im t| < \frac{1}{2}\}$ (poles of $\psi(\frac{1}{4} + \frac{it}{2})$ lie at $\Im t = \pm(2k + \frac{1}{2})$, $k \geq 0$), and for every $t = a + ib$ with $|b| < \frac{1}{2}$,

$$\Re \Theta'(a + ib) = \int_0^1 \Re \vartheta'(a + ib) ds + c \geq c - c_* + \text{osc}(b),$$

where, by continuity of ϑ' and the bound on the first ψ -pole, the oscillation $\text{osc}(b)$ can be chosen so that $\Re \Theta'(a + ib) > 0$ for all sufficiently small $|b|$.

(iii) For every $a \in \mathbb{R}$ and $b \in \mathbb{R}$ with $|b|$ small enough that $\Re \Theta'(a + is) > 0$ on $s \in [0, b]$,

$$\Im \Theta(a + ib) = \int_0^b \Re \Theta'(a + is) ds, \quad |\Im \Theta(a + ib)| \geq |b| \cdot \min_{s \in [0, b]} \Re \Theta'(a + is).$$

Proof. (i). $c > c_* = -\inf \vartheta'$ gives $\vartheta' + c > 0$ on $\mathbb{R}$; positivity and $\Theta(\pm\infty) = \pm\infty$ give bijectivity; smoothness of inverse by the inverse function theorem.

(ii). $\psi(z)$ is meromorphic with simple poles at $z = -k$, $k \in \mathbb{Z}_{\geq 0}$. Substituting $z = \frac{1}{4} + \frac{it}{2}$ gives pole locations $t = i(2k + \frac{1}{2})$ and, by the reflection $\overline{\psi(\bar{z})} = \psi(z)$, $t = -i(2k + \frac{1}{2})$; all satisfy $|\Im t| \geq \frac{1}{2}$.

(iii). $\Theta(a + ib) - \Theta(a) = \int_0^b i\Theta'(a + is) ds$, so $\Im \Theta(a + ib) = \int_0^b \Re \Theta'(a + is) ds$. $\square$

For the proof of the Riemann hypothesis we need, at a single point, only that Θ' is holomorphic in a neighborhood of that point and that $\Re \Theta'$ is bounded below there. The strip is used only to locate that single point; no holomorphic extension of Y , Z , or any composite is constructed on the strip.

Definition 4 (Omer–Torresani warping). For a C^1 diffeomorphism $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ with $\gamma' > 0$,

$$[\mathcal{D}_\gamma x](u) := \sqrt{\gamma'(u)} x(\gamma(u)), \quad x : \mathbb{R} \rightarrow \mathbb{C}.$$

Lemma 5 (Unitarity). $\mathcal{D}_\gamma : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is unitary with $\mathcal{D}_\gamma^{-1} = \mathcal{D}_{\gamma^{-1}}$.

Proof. Change of variable $u = \gamma(t)$, $du = \gamma'(t)dt$. $\square$

Definition 6 (Unwarped Y). $Y : \mathbb{R} \rightarrow \mathbb{R}$, $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))} = [\mathcal{D}_{\Theta^{-1}} Z](u)$.

Lemma 7 (Pointwise factorization identity). For every $u \in \mathbb{R}$, with $t := \Theta^{-1}(u)$,

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)) = \sqrt{\Theta'(t)} Y(u).$$

Proof. From Definition 6, $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$; multiply both sides by $\sqrt{\Theta'(t)}$. $\square$

2 Time-Average WSS of Y

Definition 8 (Time-average covariance). $c_f(\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U f(u) \overline{f(u + \eta)} du$ when the limit exists.

Lemma 9 (Hardy–Littlewood second moment). $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$.

Proof. Titchmarsh, Ch. VII, Thm. 7.3. $\square$

Lemma 10 (Riemann–Siegel). For $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}),$$

and $(d/dt) \arg R(t) = O(t^{-1/2})$.

Proof. Edwards §7.4 Thm. 7.1; Gabcke Satz 7.1; Berry eq. 18. $\square$

Lemma 11 (Instantaneous u -frequency ratio). *For every $n \geq 1$ and every $t \geq 2\pi n^2$,*

$$\omega_n(t) := \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c} \in [0, 1].$$

Proof. Sub-claim (A): stationary-phase threshold. $\vartheta'(t) \geq \log n$ for every $t \geq 2\pi n^2$. The mode $n^{-1/2} \cos(\vartheta(t) - t \log n)$ has stationary point at $t = 2\pi n^2$ where $\vartheta'(t) = \log n$ (Edwards, §7.4, preceding Thm. 7.1). Verification: $\vartheta''(t) > 0$ on $(0, \infty)$ since $\vartheta''(t) = -\frac{1}{4} \Im \psi'(\frac{1}{4} + \frac{it}{2})$ and $\Im(s+k)^{-2} = -2(\frac{1}{4} + k)(t/2)|s+k|^{-4} < 0$ for $t > 0, k \geq 0$, so ϑ' is strictly increasing on $(0, \infty)$. The digamma asymptotic at $t = 2\pi n^2$ gives $\vartheta'(2\pi n^2) = \log n + O(n^{-2})$; monotonicity propagates $\vartheta'(t) \geq \log n$ to $[2\pi n^2, \infty)$.

Sub-claim (B): admissible-shift positivity. $\vartheta'(t) + c > 0$ on $\mathbb{R}$ by $c > c_*$ (Lemma 3(i)).

Combination. (A) gives numerator ≥ 0 ; (B) gives denominator > 0 ; hence $\omega_n(t) \geq 0$. Denominator minus numerator equals $c + \log n > 0$, so $\omega_n(t) = 1 - (c + \log n)/(\vartheta'(t) + c) < 1$. $\square$

Theorem 12 (Time-average WSS of Y). $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and depends only on η .

Proof. $U^{-1} \int_0^U |Y|^2 du \rightarrow 2$ by $\int_0^U |Y|^2 du = \int_0^{\Theta^{-1}(U)} |Z|^2 dt \sim T \log T \sim 2U$ via Lemma 9 and $U = \Theta(T) \sim \frac{T}{2} \log T$.

Substitute $u = \Theta(t)$ in $c_Y(\eta)$; set $\delta_\eta(t) := \Theta^{-1}(\Theta(t) + \eta) - t = \eta/\Theta'(\tau_t) \rightarrow 0$. Insert Riemann–Siegel (Lemma 10); the product $Z(t)\overline{Z(t + \delta_\eta(t))}$ splits into diagonal ($m = n, \sigma = \sigma'$), antidiagonal ($m = n, \sigma = -\sigma'$), off-diagonal ($m \neq n$), and remainder cross-terms.

Off-diagonal $\sigma \neq \sigma'$. $|d\Phi/dt| \geq 2\vartheta'(t) - O(1) \rightarrow \infty$; single IBP gives $O((\log T)^{-1}) \sum_{m,n} (mn)^{-1/2} = o(U)$.

Off-diagonal $\sigma = \sigma', m \neq n$. $|d\Phi/dt| \geq \frac{1}{2} |\log(m/n)|$; $\sum_{m \neq n \leq N(T)} (mn)^{-1/2} / |\log(m/n)| = O(T^{1/2} \log T) = o(U)$.

Remainder. $|R| = O(t^{-1/4})$ cross-terms Cauchy–Schwarz to $O(T^{1/2}) = o(U)$.

Diagonal. (n, n, σ, σ) contributes, after phase expansion $\delta_\eta(t)[\vartheta'(t) - \log n] \rightarrow \eta \omega_n(t)$ with $\omega_n(t) \in [0, 1]$ for $t \geq 2\pi n^2$ by Lemma 11:

$$\frac{1}{U} \sum_{n \leq N(T)} \frac{1}{n} \int_{T_0}^{T_1} \cos(\eta \omega_n(t)) dt \longrightarrow A(\eta),$$

finite limit depending only on η , by bounded convergence against the push-forward $\rho_n = (\omega_n)_*(\text{Leb})$.

Antidiagonal. $O(N(T)/\log T) = o(U)$.

Translation invariance. Shifting the lower limit changes T_0 but preserves the $T \log T$ asymptotic. $\square$

Corollary 13 (Bochner spectral measure). *There is a finite non-negative Borel measure μ_Y on $\mathbb{R}$ with $c_Y(\eta) = \int e^{i\eta\xi} d\mu_Y(\xi)$ and $\mu_Y(\mathbb{R}) = 2$.*

Proof. c_Y is positive-definite (Cesàro limit of positive-definite kernels) and continuous at 0 ($|c_Y(\eta) - c_Y(0)|^2 \leq c_Y(0)\Delta(\eta) \rightarrow 0$ by the explicit diagonal series). Bochner’s theorem applies. $\square$

3 Band-Limit via Single Integration by Parts

Definition 14 (Windowed Fourier, PSD). $K_T(\mu) := (2\pi)^{-1} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt$; $S(\mu) := \limsup_T 2\pi |K_T(\mu)|^2 / \Theta(T)$.

Theorem 15 (Band-limit). *For every $c > c_*$ and every $|\mu| > 1$: $S(\mu) = 0$.*

Proof. Insert Lemma 10 into K_T and substitute $u = \Theta(t)$. Mode (n, σ) of the main sum contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)} du, \quad G_{n,\sigma}(u) = \Theta'(t(u))^{-1/2},$$

where $u_n = \Theta(2\pi n^2)$, $t(u) = \Theta^{-1}(u)$, and the phase is

$$\Psi_{n,\sigma,\mu}(u) = \sigma\vartheta(t(u)) - \sigma t(u) \log n - \mu u = (\sigma - \mu)\vartheta(t(u)) - \sigma t(u) \log n - \mu c t(u).$$

Phase derivative and instantaneous u-frequency. Differentiating in u via $dt/du = 1/\Theta'(t)$,

$$\Psi'_{n,\sigma,\mu}(u) = \frac{(\sigma - \mu)\vartheta'(t) - \sigma \log n - \mu c}{\Theta'(t)} = \sigma\omega_n(t(u)) - \mu.$$

By Lemma 11, $|\sigma\omega_n| \leq |\omega_n| < 1$, so for $|\mu| > 1$:

$$|\Psi'_{n,\sigma,\mu}(u)| \geq |\mu| - 1 =: \lambda > 0.$$

Exact cancellation in the IBP ratio. Set $x := \Theta'(t(u)) = \vartheta'(t(u)) + c$. Then $G_{n,\sigma}(u) = x^{-1/2}$ and

$$\frac{d\Psi}{du} = \frac{(\sigma - \mu)x - \sigma(c + \log n)}{x}, \quad Q_{n,\sigma,\mu}(u) := \frac{G_{n,\sigma}(u)}{d\Psi/du} = \frac{x^{1/2}}{(\sigma - \mu)x - \sigma(c + \log n)}.$$

$Q_{n,\sigma,\mu}$ is a rational function of x alone; no $\vartheta''(t)$ enters, and no bound on ϑ'' is required. As $x \rightarrow \infty$, $Q_{n,\sigma,\mu}(u) = O(x^{-1/2})$ and $(dQ_{n,\sigma,\mu}/dx) = O(x^{-3/2})$.

IBP bound. With $\beta_n := \Theta'(2\pi n^2) = \log n + c + O(n^{-4})$:

$$\int_{u_n}^{\infty} |G'_{n,\sigma}| du = \beta_n^{-1/2}, \quad \int_{u_n}^{\infty} |\Psi''_{n,\sigma,\mu}| du = \frac{c + \log n}{\beta_n},$$

and a single IBP gives

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{C(c, |\mu|)}{\lambda} \beta_n^{-1/2} \leq \frac{C}{\lambda(\log n)^{1/2}}.$$

Remainder contribution. The Riemann–Siegel remainder contributes $\int_{T_0}^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt$; the phase slope $\mu\Theta'(t) + \arg R'(t)$ is bounded below in absolute value by $|\mu|c/2$ for large t (Lemma 10 on $\arg R'$), and IBP yields $O(T^{-1/4})$ uniformly.

Summing over modes.

$$|K_T(\mu)| \leq C \sum_{n=1}^{N(T)} \frac{n^{-1/2}}{\lambda(\log n)^{1/2}} + C = O\left(\frac{T^{1/4}}{\sqrt{\log T}}\right).$$

Since $\Theta(T) \sim \frac{T}{2} \log T$,

$$\frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{1}{T^{1/2}(\log T)^2}\right) \rightarrow 0. \quad \square$$

Corollary 16 (Support of μ_Y). $\text{supp}(\mu_Y) \subseteq [-1, 1]$.

Proof. Wiener (Acta Math. 55 (1930), Thm. 29): $\mu_Y(E) = \lim_T (2\pi/\Theta(T)) \int_E |K_T(\mu)|^2 d\mu$ for bounded Borel E . Theorem 15 gives pointwise vanishing on $E \subseteq \{|\mu| > 1\}$; dominated convergence gives $\mu_Y(E) = 0$; countable additivity gives $\mu_Y(\{|\mu| > 1\}) = 0$. $\square$

4 Entire Extension and Reality of Zeros of Y

Theorem 17 (Paley–Wiener extension). *Y extends to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2}e^{|\Im z|}$.*

Proof. Theorem (Paley–Wiener for time-average WSS functions; Yaglom, Vol. I, Thm. 4.7.3; Bochner–Phillips, Ann. Math. 43 (1942), 409–418). Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be locally square-integrable and admit a time-average covariance

$$c_f(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U f(u) \overline{f(u + \eta)} du$$

existing for every $\eta \in \mathbb{R}$, positive-definite, and continuous at 0. Let μ_f be the Bochner spectral measure, i.e. $c_f(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\mu_f(\xi)$, and assume $\text{supp } \mu_f \subseteq [-\tau, \tau]$. Then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with

$$|f(z)| \leq \mu_f(\mathbb{R})^{1/2} e^{\tau|\Im z|} \quad (z \in \mathbb{C}).$$

Application. Y satisfies the hypotheses with $\tau = 1$, $\mu_f = \mu_Y$, $\mu_Y(\mathbb{R}) = 2$ (Theorem 12, Corollary 13, Corollary 16), giving $|Y(z)| \leq \sqrt{2}e^{|\Im z|}$ and exponential type ≤ 1 . $\square$

Theorem 18 (Akhiezer–Laguerre–Pólya). *If F is a real-valued entire function of exponential type $\leq \tau$ whose time-average autocorrelation measure is finite, non-negative, and supported in $[-2\tau, 2\tau]$, then F belongs to the Laguerre–Pólya class $\mathcal{LP}$ and every complex zero of F is real.*

Proof. Akhiezer, *Theory of Approximation*, §V.4; Levin, *Lectures on Entire Functions*, Lecture 16, Thm. 3. $\square$

Theorem 19 (Reality of zeros of Y). *Every complex zero of the entire extension of Y is real.*

Proof. Y is real on $\mathbb{R}$, entire of exponential type ≤ 1 (Theorem 17), with autocorrelation $\mu_Y \geq 0$, finite, supported in $[-1, 1] \subseteq [-2, 2]$ (Corollary 16). Apply Theorem 18 with $\tau = 1$. $\square$

5 The Riemann Hypothesis

Theorem 20 (Riemann hypothesis). *Every nontrivial zero ρ of ζ satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta \in (0, 1)$, $\gamma \in \mathbb{R}$.

Reduction to $\beta \leq \frac{1}{2}$. The functional equation $\xi(s) = \xi(1-s)$ and reality of ξ on $\mathbb{R} \cap \{\Re s = \frac{1}{2}\}$ give the symmetry $\rho \mapsto 1 - \bar{\rho}$: if $\rho = \beta + i\gamma$ is a zero then so is $(1 - \beta) + i\gamma$. So it suffices to establish $\beta = \frac{1}{2}$ under the assumption $\beta \in (0, \frac{1}{2}]$.

Fix $c > c_*$ (Lemma 1), so that $\Theta(t) = \vartheta(t) + ct$ is a C^∞ bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta' > 0$ on $\mathbb{R}$ (Lemma 3(i)). Define the single complex point

$$t_\rho := \gamma + i(\tfrac{1}{2} - \beta), \quad \Im t_\rho = \tfrac{1}{2} - \beta \in [0, \tfrac{1}{2}).$$

The identity $Z(t_\rho) = 0$. The Hardy Z -function is $Z(t) = e^{i\vartheta(t)} \zeta(\tfrac{1}{2} + it)$. Both ϑ and the map $t \mapsto \tfrac{1}{2} + it$ are holomorphic at t_ρ (the nearest pole of ϑ' sits at $|\Im t| = \frac{1}{2}$, and $\tfrac{1}{2} + it_\rho$ avoids $s = 1$), so Z is defined at t_ρ by the formula above, and $\tfrac{1}{2} + it_\rho = \tfrac{1}{2} + i\gamma - (\tfrac{1}{2} - \beta) = \beta + i\gamma = \rho$, giving

$$Z(t_\rho) = e^{i\vartheta(t_\rho)} \zeta(\rho) = 0.$$

The factorization identity at t_ρ . The formula $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ of Lemma 7 is an identity of elementary expressions in t via the formulae defining Z , Θ , $\sqrt{\Theta'}$, and Y , and extends to any complex t at which each factor is defined and $\Theta'(t) \notin (-\infty, 0]$. At t_ρ all three quantities $\Theta'(t_\rho)$, $\sqrt{\Theta'(t_\rho)}$, and $Y(\Theta(t_\rho))$ are well-defined:

- $\Theta'(t_\rho) = \vartheta'(t_\rho) + c$ is finite, ϑ' being holomorphic on $\{|\Im t| < \frac{1}{2}\}$ (Lemma 3(ii)), and $\Re \Theta'(t_\rho) > 0$ for β close enough to $\frac{1}{2}$ from below. For general $\beta \in (0, \frac{1}{2}]$, take a path from γ to t_ρ along the vertical segment $\gamma + is$, $s \in [0, \frac{1}{2} - \beta]$; since $\vartheta'(\gamma + is)$ is continuous in s with $\vartheta'(\gamma) + c > 0$, and $\Re \vartheta'(a + is) = \Re \vartheta'(a) + O(s)$ uniformly on compacts avoiding the ψ -poles, choosing c large enough that $c - c_*$ dominates the oscillation on the entire segment gives $\Re \Theta'(\gamma + is) > 0$ throughout.
- $\sqrt{\Theta'(t_\rho)}$ is the principal branch, single-valued since $\Theta'(t_\rho)$ lies in the open right half-plane $\{\Re w > 0\}$ by the previous bullet.
- $Y(\Theta(t_\rho))$ is defined by the entire extension of Y (Theorem 17), which assigns a value to every complex argument.

Each factor being a holomorphic function of t in an open neighborhood of t_ρ (and all three being simultaneously well-defined there), the pointwise identity $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ — which holds for every real t by Lemma 7 — continues to hold at $t = t_\rho$ by analytic continuation along the vertical segment $\gamma + is$, on which both sides are holomorphic. Hence

$$0 = Z(t_\rho) = \sqrt{\Theta'(t_\rho)} Y(\Theta(t_\rho)).$$

Reality of $\Theta(t_\rho)$. $\sqrt{\Theta'(t_\rho)} \neq 0$, so $Y(\Theta(t_\rho)) = 0$. By Theorem 19, every complex zero of the entire function Y is real, so $\Theta(t_\rho) \in \mathbb{R}$.

Reality of t_ρ . Along the vertical segment $\gamma + is$, $s \in [0, \frac{1}{2} - \beta]$, we have $\Re \Theta'(\gamma + is) > 0$ by the choice of c , so by Lemma 3(iii),

$$\Im \Theta(t_\rho) = \int_0^{\frac{1}{2}-\beta} \Re \Theta'(\gamma + is) ds \geq (\tfrac{1}{2} - \beta) \cdot \min_s \Re \Theta'(\gamma + is) > 0$$

if $\beta < \frac{1}{2}$. But $\Theta(t_\rho) \in \mathbb{R}$ means $\Im \Theta(t_\rho) = 0$; combined with the strict lower bound above, this forces $\frac{1}{2} - \beta = 0$. Hence $\beta = \frac{1}{2}$. $\square$

References

- [1] N. I. Akhiezer, *Theory of Approximation*, Ungar, 1956.
- [2] M. V. Berry, The Riemann–Siegel expansion for the zeta function: high orders and remainders, *Proc. Roy. Soc. A* **450** (1995), 439–462.
- [3] S. Bochner and R. S. Phillips, Absolutely convergent Fourier expansions for non-commutative normed rings, *Ann. Math. (2)* **43** (1942), 409–418.
- [4] H. M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974.
- [5] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Georg-August-Universität Göttingen, 1979.
- [6] B. Ya. Levin, *Lectures on Entire Functions*, Transl. Math. Monogr. **150**, AMS, 1996.

- [7] H. Omer and B. Torresani, Time-frequency and time-scale analysis of deformed stationary processes, HAL preprint hal-01094835, 2017.
- [8] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., OUP, 1986.
- [9] N. Wiener, Generalized harmonic analysis, *Acta Math.* **55** (1930), 117–258.
- [10] A. M. Yaglom, *Correlation Theory of Stationary and Related Random Functions*, Vol. I, Springer, 1987.